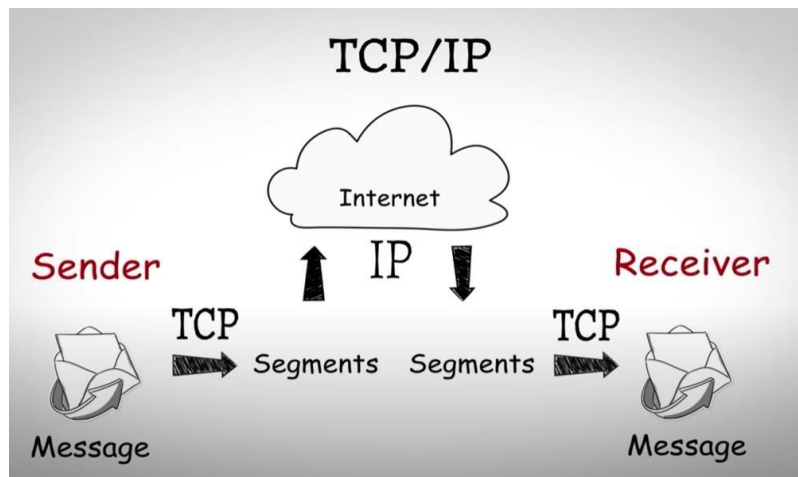


TCP/IP (Transmission Control Protocol / Internet Protocol)- TCP

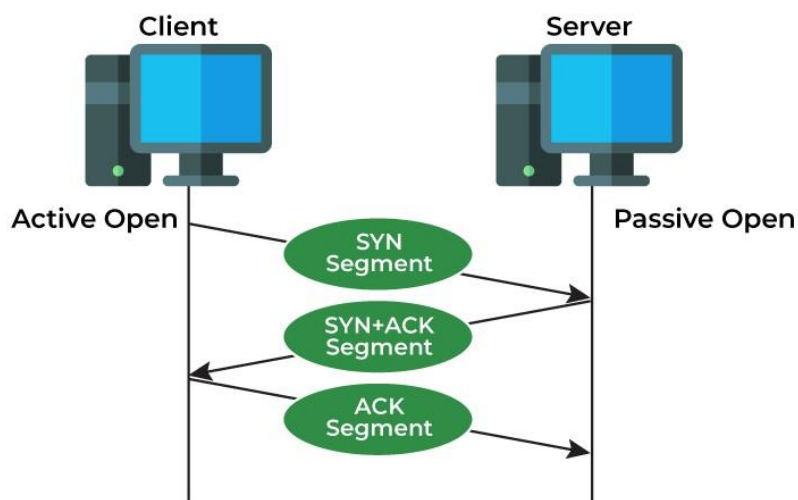
break down message into segments at sender end.

IP is use for routing these segments to destination through ipv4 address in Internet. TCP reassemble these segments into message at receiver end.

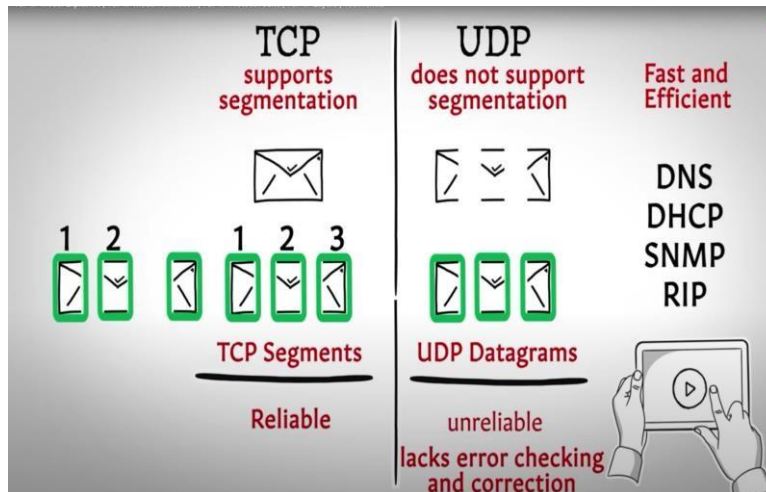


TCP three way handshaking

Before transmitting data, first Client send **SYN segment** (initial request) and then server reply with **SYN + ACK packet**, client then re send **acknowledgement packet** to receiver. After receiving ack packet then connection is established.



TCP vs Difference -



TCP

UDP

- | | |
|--|------------------------------------|
| 1. TCP is connection-oriented protocol | UDP is not connection-oriented |
| 2. TCP is reliable as it guarantees of data to the destination | UDP is not reliable delivery |
| 3. TCP connection is a byte stream. | UDP connection is a message stream |
| 4. TCP is slower than UDP | UDP speed is higher than TCP |
| 5. TCP manages sequence order of segments | UDP does not maintain data |
| 6. Flow control & error control | UDP has no flow control & EC |

TCP/IP MODEL

The TCP/IP model, also known as the Internet protocol suite, is a conceptual framework for how data is transmitted across networks, especially the internet. It defines a set of rules (protocols) that govern communication between devices, enabling them to exchange information seamlessly. The model is structured into four layers, each with specific functions: Application, Transport, Internet, and Network Access

Protocols at Each OSI Layer

Application Layer – HTTP, HTTPS, FTP, SFTP, SMTP, POP3, IMAP, SSH, Telnet

Presentation Layer – SSL/TLS, JPEG, MPEG

Session Layer – NetBIOS, RPC

Transport Layer – TCP, UDP

Network Layer- IP (V4, V6), ICMP, SNMP, IPSec

Data Link Layer – ARP, PPP, Ethernet

Physical Layer – Cable, NICs, Hub

NetBIOS is an API (Application Programming Interface) that allows applications on different computers to communicate over a local area network (LAN). It used in Windows-based networks

Key Features:

Name Service: Allows applications to register and resolve names to IP addresses. Each device on the network has a NetBIOS name (up to 16 characters).

Session Service: Enables two computers to establish a connection and communicate in a session-oriented manner.

Datagram Service: Allows connectionless communication, useful for broadcasting messages.

Use Cases:

File and printer sharing in Windows networks.

LAN-based communication in small networks

RPC

RPC is a protocol that allows a program to execute a procedure (subroutine) on another computer as if it were local

Client-Server Model: The client remotely invokes a procedure on the server.

Feature	NetBIOS	RPC
Purpose	LAN communication	Distributed computing
Protocol Type	API over network protocols	Protocol for remote function calls
Common Use	File sharing, name resolution	Remote service execution
Transport Protocols	UDP/TCP	TCP/HTTP
Scalability	Limited	High

